Daniel E. Ripka

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Summary

Embedded Systems Engineer with 4+ years of professional experience developing firmware and software, specializing in C/C++, real-time systems, and hardware-software integration.

Skills

- Programming: C, C++, Python, Java
- Embedded Systems: FreeRTOS, CppUTest, UART, SPI, Hardware Abstraction Layers (HAL)
- Tools: Git, SVN, Bash, PowerShell, LTspice, Oscilloscopes, PicoScope, Confluence
- Development Environments: Visual Studio Code, Microchip Studio, Renesas HEW
- Data Analysis: Pandas, NumPy, Matplotlib

Experience

Badger Meter - Brown Deer, WI

Firmware Engineer, from June 2023 to Present

- Engineered a custom, interrupt-driven UART protocol for Orion Cellular Endpoints to communicate with smart water meters over a 3-wire interface, improving communication speed and power efficiency through an extensible design. Earned the 2025 1st Place Technical Achievement Award.
- Designed FreeRTOS task scheduling algorithms to optimize real-time execution, data reporting, and resource management.
- Developed CppUTest unit tests and redesigned the test framework to mock hardware peripherals and states, improving firmware validation.
- Developed hardware abstraction layers (HAL), device drivers, and UART/SPI interfaces for embedded peripherals.
- Created Python automation scripts for firmware testing and device configuration.
- Analyzed AWS-sourced field device data using Pandas and Matplotlib.
- Conducted environmental testing, including battery current and power consumption measurements.

Columbus McKinnon Corporation, Magnetek - Menomonee Falls, WI

Associate Engineer, from July 2021 to June 2023

- Developed VFD firmware simulating crane loads on a hoist using advanced motor dynamics.
- Engineered overhead crane slowdown firmware triggered by limit switches, maximizing operating space while maintaining safety, with integrated laser device support.
- Developed PyQt desktop applications for VFD parameter testing and monitoring using Modbus Serial Protocol.

Education

Bachelor of Science in Computer Engineering
University of Wisconsin - Milwaukee